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Super Resolution Enhanced Hybrid Image Log Unsupervised Facies Classification in Core Samples

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Abstract

Automated lithofacies classification from core images can reduce subjectivity and speed up geological interpretation. This study introduces an advanced workflow for lithofacies classification using machine learning applied to core images from the Pohokura wells in New Zealand. The work was driven by three key goals: enhancing image resolution, segmenting geological features, and improving lithofacies classification accuracy.

The first step involved resolution enhancement using the Enhanced Super-Resolution Generative Adversarial Network (ESRGAN), trained on high-resolution core images from the Wisting Field in Norway. The ESRGAN algorithm effectively increased image resolution by fourfold, allowing for detailed feature extraction critical for accurate segmentation. Following this, the U-Net architecture was employed for image segmentation, where manually labeled patches from enhanced white and UV light core images served as the training dataset. The U-Net was further optimized by combining both light sources as input channels, resulting in an input size of 50x50x6 and improved segmentation outcomes.

The final step involved a majority voting mechanism, where segmented patches were reassembled into their original depth positions, and the dominant lithofacies category at each depth was assigned. This combined approach yielded a facies prediction accuracy of 77%, significantly outperforming the 2023 study by Ingrid Gjerazi, which utilized the same dataset but achieved lower prediction accuracy due to computational and methodological limitations.

The key contributions of this study include overcoming data quality and imbalance challenges by leveraging ESRGAN for resolution enhancement and by expanding the dataset with white and UV images; implementing a more robust image segmentation and classification framework, achieving higher predictive accuracy; and delivering a workflow that is not only reproducible but also scalable for future studies involving core image datasets.

This work demonstrates the potential of combining advanced machine learning techniques with high-resolution image processing to achieve precise geological interpretations, paving the way for similar applications in subsurface data analysis.

Objectives

In this study, we aimed to leverage machine learning techniques for lithofacies classification in the Pohokura wells by analyzing core images. The specific objectives were as follows:

- **Resolution Enhancement:** To enhance the resolution of low-resolution white and ultraviolet (UV) core images from Pohokura by using high-resolution core images from another field as reference data.
- **Image Segmentation:** To segment these enhanced-resolution core images, delineating distinct geological features with greater precision and detail.
- **Lithofacies Prediction:** To implement a voting-based classification mechanism that aggregates segmented data to assign lithofacies categories across specific depth intervals.

These objectives were achieved through the integration of enhanced resolution imaging, deep learning-based segmentation, and systematic classification, resulting in superior facies prediction accuracy and addressing limitations observed in earlier studies.

PREVIOUS STUDIES AND APPROACH

Several previous studies laid the groundwork for this project by exploring the use of machine learning techniques for lithofacies classification using core images. Two significant studies conducted in 2021 and 2022 have provided critical insights and methodologies that shaped our approach.

The [2021 study, led by Jörg Peisker](#), used well data from Norway to develop a deep learning neural network specifically designed for lithofacies characterization. Peisker's work involved creating a Python script that employed convolutional neural network (CNN) architectures to identify lithofacies features such as bedding and textures directly from the core images. The main conclusion of this study was that deep learning could effectively capture key lithological features, but there were limitations in computational efficiency, as training the model required significant time and resources.

Building on Peisker's work, the 2023 study led by Ingrid Gjerazi focused on well data from three New Zealand wells, including the Pohokura wells. Gjerazi utilized the neural network code developed by Peisker but faced several challenges in the implementation phase. The main obstacle was setting up the Python environment due to IT security restrictions, which blocked many of the necessary installations. To circumvent these issues, Gjerazi opted to use Dataiku, the data science platform available within the Delfi environment. However, at that time, Dataiku was not fully mature for handling complex machine learning models, which led to limitations such as reduced training epochs and lower accuracy in the predictions.

Gjerazi's work employed two deep neural networks—ResNet50 and GAP CNN—for achieving high-resolution image segmentation and lithofacies classification. The underlying framework for these models was TensorFlow. High-resolution images from another field were used as reference data to enhance the lower resolution images from Pohokura. Image augmentation techniques—including rotation, resizing, shifting, and contrast adjustments—were applied to expand the dataset and address class imbalance, a recurring issue in lithofacies classification tasks.

The segmentation process involved delineating geological features from these enhanced images, followed by integrating well log data to improve classification accuracy. Gjerazi concluded that the use of high-resolution images significantly enhanced the model's ability to identify lithological features. Despite the promising results, the study was limited by the maturity of the tools available at the time, as well as by computational restrictions that prevented a thorough exploration of the models' capabilities. The project faced several challenges, including data imbalance, poor data quality, environment setup difficulties, and computational limitations. Data imbalance resulted in overtraining of certain dominant lithofacies classes, while the quality of Pohokura core images was generally lower compared to reference datasets, impacting training precision. Environment setup also posed significant challenges, especially in configuring the

necessary Python packages due to IT restrictions. Using Dataiku helped mitigate some of these challenges, although memory constraints led to frequent session crashes during extended training periods.

Furthermore, computing resources posed a considerable obstacle. Training the machine learning model required substantial computational power, and the high volume of core images (~16,000) generated during preprocessing frequently led to crashes. Despite these issues, we managed to improve upon the previous results, thanks to careful management of epochs, data balancing techniques, and utilizing available computational infrastructure efficiently. One of the main challenges for this project was data imbalance. For the more dominant lithofacies classes, there was a large number of core images, which resulted in overtraining the model. This imbalance posed the risk of predicting more of one facies and not adequately training for the underrepresented classes. Additionally, the quality of the core images from the Pohokura wells was lower, reducing the likelihood of improved training and prediction outcomes. Another major challenge faced was the setup of the Python environment. Due to IT Security restrictions, many required installations were blocked, and persistent errors led to the decision to use the Dataiku platform. Despite its versatility, Dataiku also presented limitations, including frequent crashes during model training with larger numbers of epochs due to memory constraints.

The 2023 iteration of this study aimed to address the limitations observed in the earlier work, particularly by increasing the number of training epochs and enhancing the computational setup to allow for deeper learning and more reliable predictions. By building on the foundations established by Peisker and Gjerazi, the present work seeks to overcome previous limitations and further refine the methodology for lithofacies classification in the Pohokura wells.

Data Availability

Three Pohokura wells, Pohokura 1 (POH-01), Pohokura 2 (POH-02), and Pohokura South 1 (POS-1), were made available for this project, with a total of 176 meters of core imaged under both white and UV light conditions. The location of the wells is shown in [Figure 1](#). In total, 27 facies classes were identified, and these were used to train the machine learning model. [Table 1](#) shows the file size and number of images of the information used. Specifically, the images from the three wells varied in terms of resolution, file size, and number of images. For Pohokura 1, core images were taken between 3527 and 3553 meters depth, with file sizes ranging from 70 to 125 KB, at a resolution of 884×1156 pixels, and stored in .jpg format. A total of 5 images were used for this well. Pohokura 2 provided core images from depths of 3575 to 3629 meters, with a slightly broader range of file sizes, from 46 to 148 KB, and the same resolution of 884×1156 pixels, also in .jpg format, resulting in 10 images. Pohokura South 1, on the other hand, covered depths from 3680 to 3773 meters and had much larger file sizes, ranging from 4 to 6 MB, with a higher resolution of 2034×2660 pixels, also in .jpg format, contributing 16 images. This indicates that the images from Pohokura South 1 have the highest resolution, which likely provided more detailed geological information compared to the other wells. Images were sliced into 1 cm segments and preprocessed to ensure accurate lithofacies representation before being used in training.

Table 1—Original images details for available wells.

Well	File Size	Width	Height	Format	# Images
POH-01	70 - 125 KB	884	1156	.jpg	5
POH-02	46 - 148 KB	884	1156	.jpg	10
POS-01	4 - 6 MB	2034	2660	.jpg	16

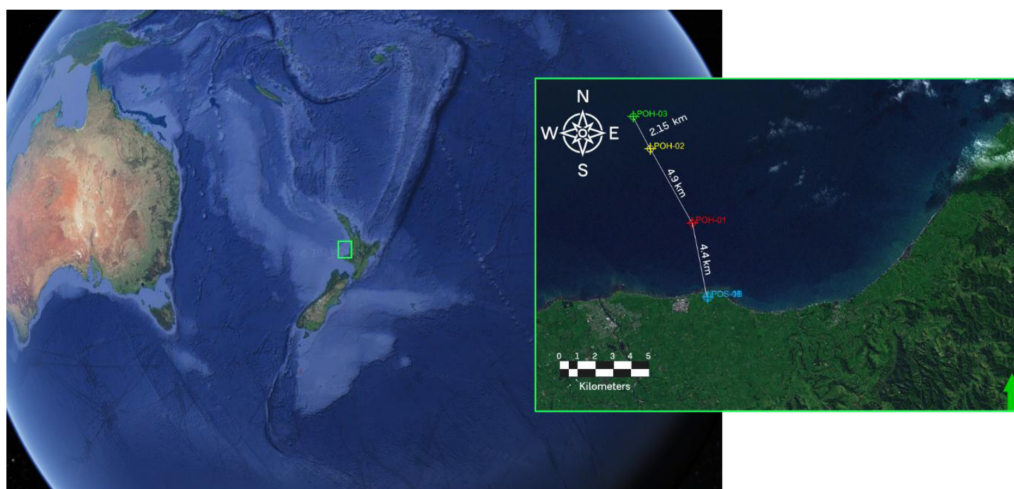


Figure 1—Area of study highlighted by green square in world map, well locations and distances for available wells in the study.

Facies Description

The following is a summary of the key facies identified from the Pohokura wells based on their lithological characteristics:

The Co (Coal) facies refer to well-developed coal units ranging from 8 cm to 65 cm in thickness, observed only in Pohokura South-01. These units are interbedded with laminated claystones, heterolithics, and bioturbated sandstones, indicating accumulation under subaerial and subaqueous conditions. In the case of Cl (Laminated Claystones), these are non-bioturbated, well-laminated claystones with occasional rootlets and plant material, commonly found in units between 2 cm to 60 cm thick. These claystones are typically interbedded with carbonaceous mudstones and coals, indicating low-energy deposition with periodic or sustained oxygen depletion.

The Mcarb (Carbonaceous Mudstone) facies consists of reddish-brown and dark-grey mudstones containing rootlets, found interbedded with coal units and heterolithics in Pohokura South-01. The presence of coal horizons suggests an environment with alternating subaerial and subaqueous conditions. The Sil (Laminated Siltstone) facies refers to siltstones with interbedded sand-rich and clay-rich facies, exhibiting sparse bioturbation. These units, ranging from 10 cm to 60 cm in thickness, represent deposition under fluctuating low-energy conditions, with occasional high-energy events resulting in ripple laminations. Hm (Muddy Heterolithics) are composed of alternating claystone and siltstone beds with fine sandstone, ranging from 20 cm to 1.8 m thick. These heterolithics show varied degrees of bioturbation and represent low-energy conditions punctuated by high-energy episodic events.

In the case of Hs (Sandy Heterolithics), these are sandstone beds interbedded with laminated claystones and siltstones, forming units up to 3 m thick. These heterolithics suggest low-energy mud deposition interrupted by episodic high-energy conditions, such as ripple laminations and hummocky cross-stratifications. The Hsb (Bioturbated Sandy Heterolithics) facies are sandy heterolithics with moderate degrees of bioturbation. Units range from 20 cm to 3 m thick and indicate depositional settings characterized by fluctuating energy levels and occasional colonization by burrowing organisms. Cl (Laminated Claystones) are non-bioturbated, well-laminated claystones with occasional rootlets and plant material, commonly found in units between 2 cm to 60 cm thick. These claystones are typically interbedded with carbonaceous mudstones and coals, indicating low-energy deposition with periodic or sustained oxygen depletion. Mcarb (Carbonaceous Mudstone) consists of reddish-brown and dark-grey mudstones containing rootlets, found interbedded with coal units and heterolithics in Pohokura South-01. The presence of coal horizons suggests an environment with alternating subaerial and subaqueous conditions. Sil (Laminated

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This facies information provides a detailed understanding of the depositional environments and lithological variability across the Pohokura wells, which is essential for supporting lithofacies classification in machine learning models. This facies information provides a detailed understanding of the depositional environments and lithological variability across the Pohokura wells, which is essential for supporting lithofacies classification in machine learning models.

Machine Learning

Histogram Segmentation

The Histogram Segmentation approach involves analyzing an image's pixel intensity distribution to partition it into distinct segments based on specific criteria, such as color or intensity ranges. This method starts by calculating the color histogram of the image, which represents the frequency distribution of color intensities for each channel (red, green, and blue). The histogram is then divided into specific sections, each corresponding to a predefined color range. Using this segmentation, portions of the original image are assigned to these sections based on their pixel values, allowing for a visualization of where specific color ranges dominate.

Figure 2 demonstrates this process using an input image from the POS-1 well. The original image is a vertical depiction of what appears to be a core sample. Its color properties have been analyzed, and the histogram has been divided into four sections: red, green, blue, and yellow. At the bottom of the figure, a histogram plot illustrates the frequency distribution of pixel intensities, with different shaded regions corresponding to the four color ranges.

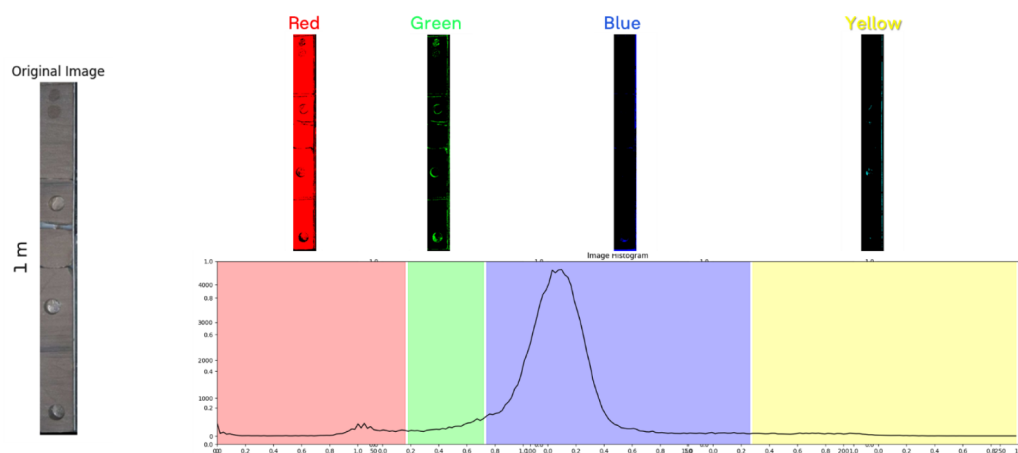


Figure 2—Histogram Segmentation for one image from well POS-1.

Above the histogram plot, reconstructed images represent the portions of the original image that belong to each color section. The reconstruction for the red section shows the most detail and closely resembles the original image, indicating that the majority of the pixel intensities in the original image fall within the red range. The green and blue reconstructions highlight only specific, less prominent regions with green or blue intensities. The yellow section reconstruction contains very sparse information, showing that this range is minimal in the original image. While the individual reconstructions provide insight into the distribution of specific color intensities, they do not fully replicate the original image due to the overwhelming dominance of the red color range in the original histogram segmentation.

K-Nearest Neighbors

K-Nearest Neighbors (K-NN) for image segmentation is a machine learning technique used to partition an image into clusters based on pixel intensity or color similarity. The process involves treating each pixel as a data point in a multidimensional space (e.g., RGB color space) and assigning it to a cluster based on the proximity of its features to predefined centroids. The number of clusters, k , determines the level of segmentation and is usually optimized using techniques like the Elbow method, which identifies the k value where the within-cluster sum of squares (WCSS) significantly decreases, representing an optimal balance between detail and simplicity.

In Figure 3, the original image, representing a core sample from the POS-1 well, was segmented using the K-NN method. The Elbow method was applied to determine the optimal number of clusters, which was found to be $k = 5$. The segmented image (Clustered Image) shows the five distinct clusters assigned by K-NN, with each cluster represented by a unique color.

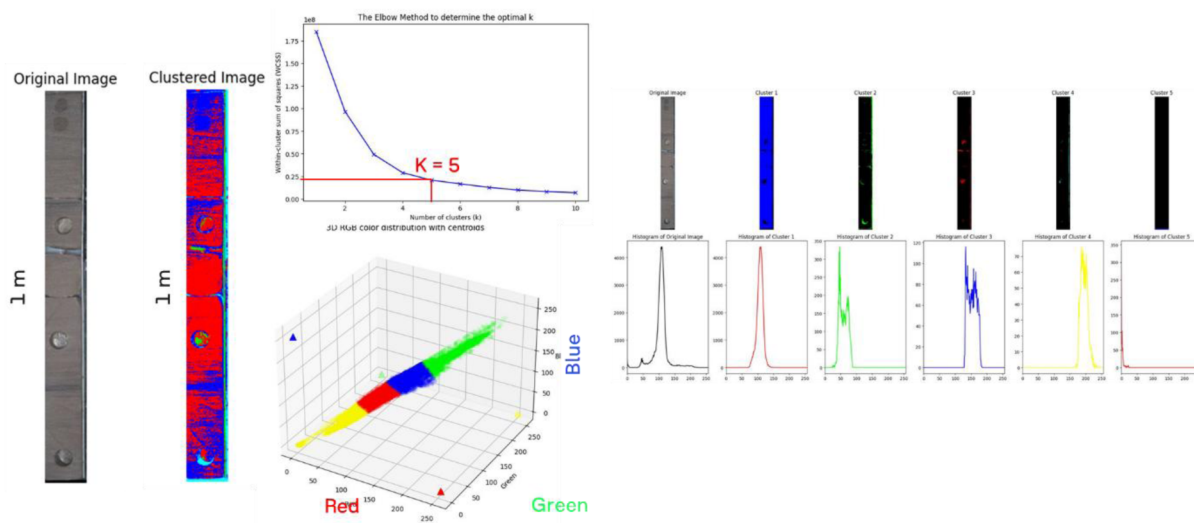


Figure 3—K-Nearest Neighbors (K-NN for one image from well POS-1.

The individual histograms for each cluster highlight the distribution of pixel intensities within each segmented region. Cluster 1 corresponds to pixels with a predominantly blue intensity, Cluster 2 emphasizes the green range, Cluster 3 captures red-dominant regions, Cluster 4 highlights yellow intensities, and Cluster 5 comprises the remaining, less significant, red-dominated values. The histograms indicate clear distinctions in intensity distributions for each cluster, offering insights into the variability of the image's color composition.

The analysis also included a 3D representation of the RGB color distribution, where each pixel's color was plotted in a three-dimensional space with axes for red, green, and blue intensities. The 3D scatter plot visualizes how the clusters are distributed across the RGB color space, with clear separations between the clusters, reinforcing the segmentation results.

After reconstructing the image based on the K-NN clusters, the segmentation results showed significant improvement over the previously described Histogram Segmentation. The K-NN method provided a more refined and detailed partitioning of the image. However, despite these advancements, the technique was insufficient to clearly classify facies using this approach, as the clusters did not directly correspond to geological or lithological features in the core sample. This suggests that while K-NN segmentation enhances color-based partitioning, additional contextual or feature-based analysis would be required for facies classification.

DEEP NEURAL NETWORK

Data and Pre-processing and Workflow. Since the previous approaches—Histogram Segmentation and K-NN—did not yield satisfactory results for classifying facies, I implemented a Deep Neural Network (DNN) to address the limitations. The first step in this workflow involved preparing the dataset by splitting the original core images into smaller, uniform segments measuring 1 meter by 8 centimeters. Figure 4 illustrates how the original images were divided into these individual sections.

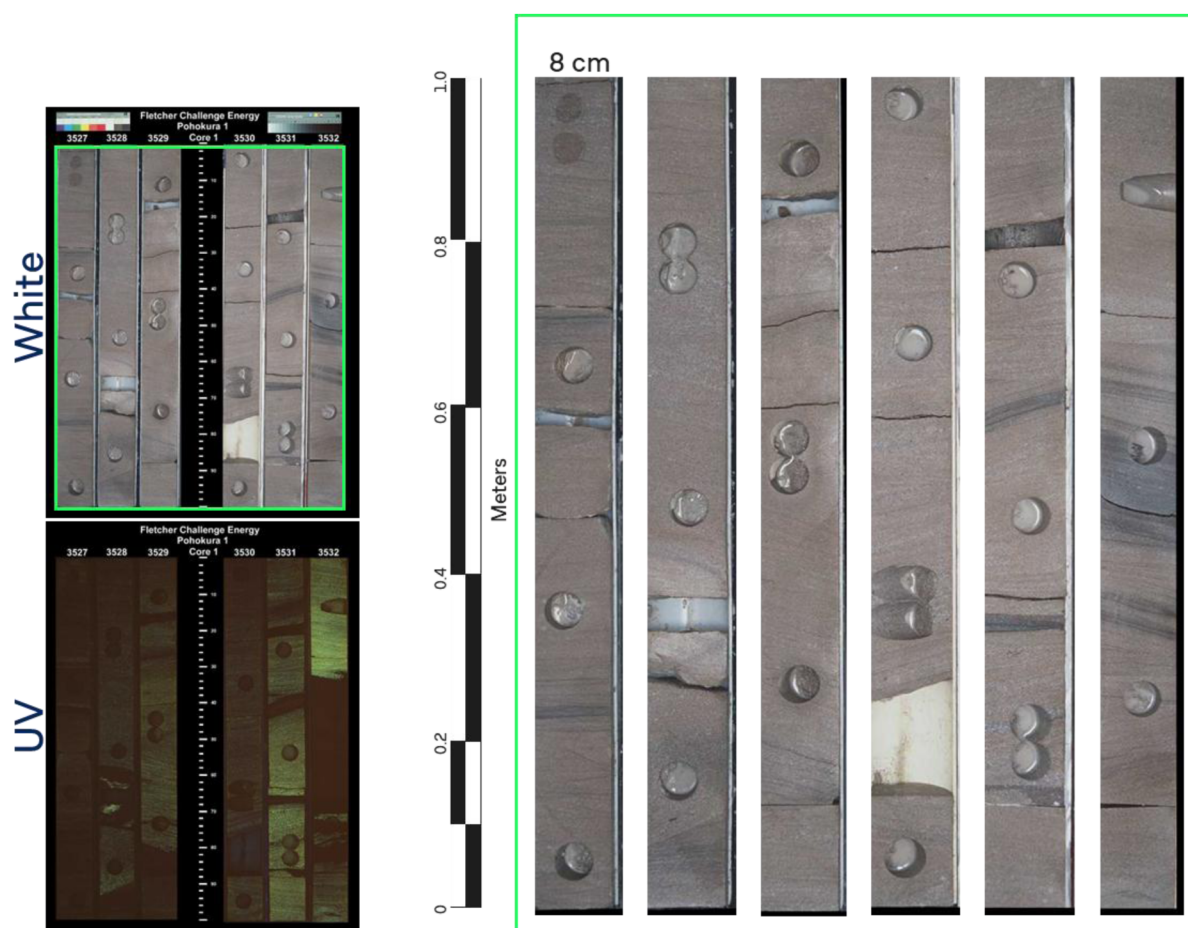


Figure 4—Well cropping summary.

This preprocessing resulted in 27 images for the POH-01 well, with approximate dimensions of 107×1021 pixels, and 55 images for the POH-02 well, with dimensions of 104×1023 pixels. For the POS-01 well, the splitting yielded 94 images, each approximately 240×2346 pixels. The histograms in Figure 5 represent the distributions of sizes, widths, and heights of the segmented images. It is evident that the POS-01 well images have significantly higher height and width compared to those from POH-01 and POH-02. This

difference is due to the higher resolution of the original images for POS-01, as shown in Table 1. The table highlights the differences in file sizes, dimensions, and the number of images per well, emphasizing the enhanced detail in the POS-01 dataset.

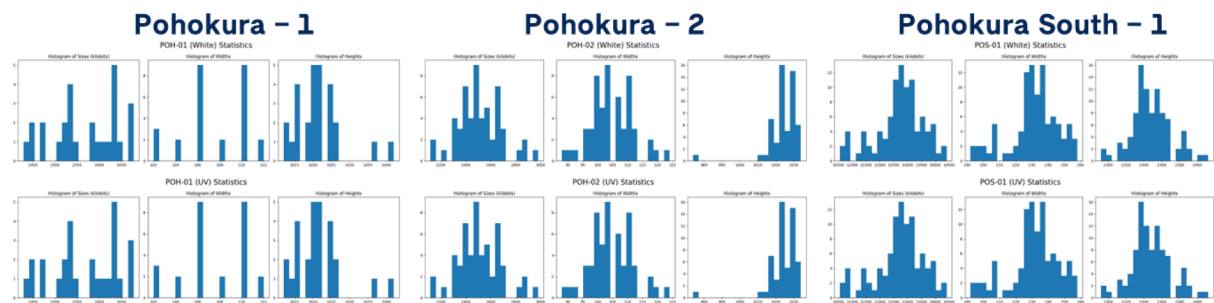


Figure 5—Histograms showing the average width and height of White and UV images after cropping.

Table 2—Image details after cropping.

Well	File Size	Width*	Height*	Format	# Images
POH-01 (White UV)	2569 KB	~ 107	~ 1021	.jpg	27
POH-02 (White UV)	40 - 60 KB	~ 104	~ 1023	.jpg	55
POS-01 (White UV)	200 - 330 KB	~ 240	~ 2346	.jpg	94

This step of segmentation and preparation ensured uniformity across the dataset while maintaining the resolution required for deep learning-based analysis, providing a solid foundation for training the Deep Neural Network.

After the images were cropped into smaller pieces, they were carefully labeled to prepare the dataset for instance segmentation. This labeling step involved manually annotating each image to assign specific categories to distinguish different elements within the core samples. These labeled images served as the foundation for training the instance segmentation model, ensuring that the neural network could accurately identify and differentiate between various features in the core samples.

In Figure 6, the differences between the approaches of "classification and location," "object detection," and "instance segmentation" are demonstrated. In the "classification and location" approach, the model identifies the class of the entire cropped image, labeling it as "bioturbated sandstone" without providing further spatial details.

In contrast, the "object detection" approach goes a step further by not only classifying the image but also locating specific objects within it. In this example, the red box highlights the presence of a circular plug, a feature extracted from the core, while the orange box represents the "undefined/no core" section, which indicates parts of the image where no core sample is present.

Finally, the "instance segmentation" process takes the analysis to a more detailed level. During the manual labeling phase, each pixel of the image was assigned to one of three instances: bioturbated sandstones, circular plug, or the "undefined/no core" section. This pixel-wise segmentation ensures a more granular understanding of the image's content, enabling the model to distinguish specific regions and features within the core sample with high precision. For the example in Figure 6, the labeled segments allow for detailed training and improved performance of the deep learning model in recognizing facies and associated features.

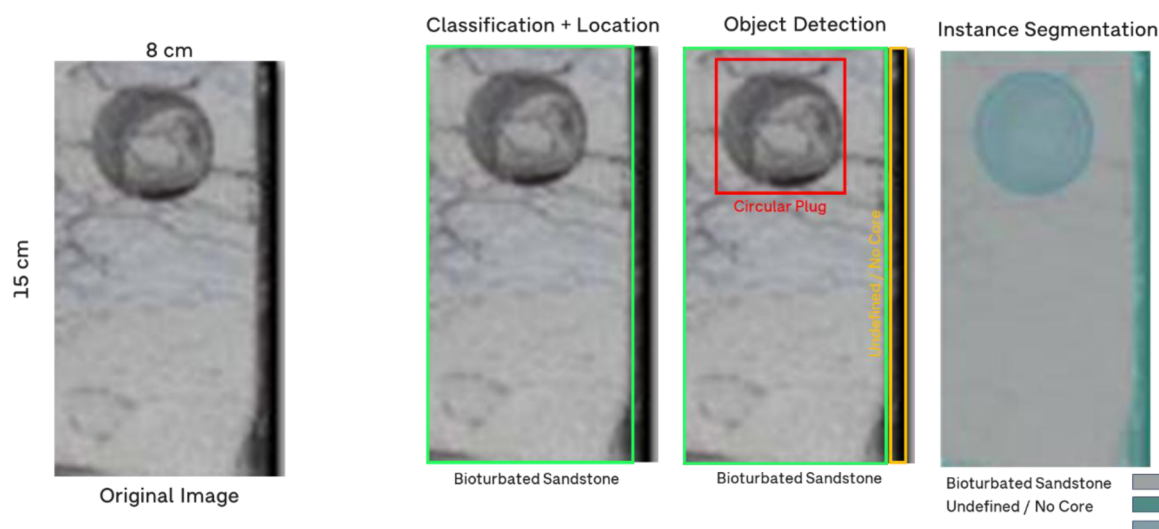


Figure 6—Comparison of classification and location, object detection, and instance segmentation approaches, showing manual labeling of bioturbated sandstone, circular plug, and "undefined/no core" sections.

Resolution Enhancement: GAN's (ESRGAN)

ESRGAN is a generative adversarial network that reconstructs high-resolution images from low-resolution inputs. We followed the training strategy of Wang et al. (2018). Low-resolution patches were generated by bicubic down-sampling high-resolution Wisting core images; training used a mini-batch of 16 and PSNR-oriented pre-training with an L1 loss to stabilise the generator [OBJ]. The generator was then fine-tuned using adversarial and perceptual losses, and Adam optimisation (learning rate 1×10^{-4} with step-wise decay) [OBJ]. Two generator variants were tested: 16 residual blocks (SRGAN-equivalent) and 23 residual-in-residual dense blocks. After training, the ESRGAN increased the spatial resolution of Pohokura images by a factor of four (e.g., $200 \times 2,300$ pixels became $800 \times 9,200$ pixels), revealing subtle sedimentary structures that were indistinguishable in the original images.

Figure 7 illustrates the workflow of the ESRGAN algorithm used in this study. The process starts with a low-resolution core image input, measuring 200×2300 pixels. This image is first passed through convolutional layers for feature extraction, followed by a series of basic blocks that correspond to the residual blocks in ESRGAN. These blocks are designed to capture fine details and enhance feature representation. After feature extraction, the network performs upsampling to increase the resolution, followed by additional convolutional layers that refine the details of the image. The final output is a high-resolution image with dimensions of 800×9200 pixels, representing a significant improvement in clarity and resolution.

The architecture includes dense blocks, which are highlighted in the zoomed-in section of Figure 7. These blocks connect multiple convolutional layers, allowing for feature reuse and improving gradient flow during training. The dense connections, combined with Leaky ReLU activation functions, contribute to the network's ability to learn and enhance intricate details in the images.

To tailor the ESRGAN algorithm for well core images, it was trained on a high-resolution dataset from the Wisting Field in Norway. This dataset comprised three wells: Central Wisting, Hansen, and Mercury. The Central Wisting dataset included 200 images with a resolution of 3034×18000 pixels. The Hansen dataset consisted of 148 images with dimensions of 1391×9000 pixels, while the Mercury dataset featured 56 images at 992×4500 pixels. Each well included both White and UV light images of core samples. Training the model on this dataset enabled it to learn specific patterns and textures unique to geological cores, allowing for high-quality resolution enhancement that preserved critical geological features. The

customized ESRGAN model provided outputs that were not only visually superior but also detailed enough for accurate geological interpretation and facies classification.

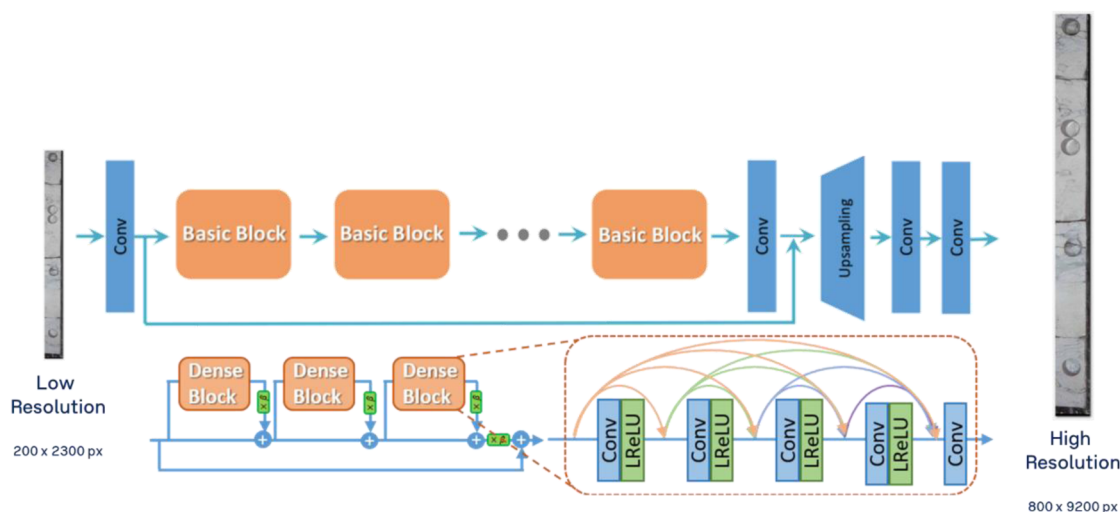


Figure 7—Workflow of the ESRGAN algorithm for resolution enhancement, converting a low-resolution core image (200 × 2300 pixels) into a high-resolution output (800 × 9200 pixels). The model incorporates convolutional layers, basic blocks, upsampling, and dense connections to refine details. The ESRGAN was trained on high-resolution core images from the Wisting Field in Norway to achieve optimal results.

Image Enhancement Results

Figure 9 showcases two images of well cores from POH-01 and POS-01 before resolution enhancement. The core image from POH-01 has a width of 8 cm, corresponding to 110 pixels, and a vertical height of 1 meter, represented by 1020 pixels. In comparison, the POS-01 image also covers a width of 8 cm but with 244 pixels and a vertical height of 1 meter represented by 2388 pixels. These measurements reflect the lower resolution of the POH-01 dataset compared to the higher resolution of the POS-01 dataset, highlighting the variability in image quality prior to enhancement. The size difference is also apparent, with POH-01 having smaller dimensions and a file size of 49 KB compared to the larger and more detailed POS-01 file at 276 KB. This variability emphasizes the importance of resolution enhancement techniques for creating consistent, high-quality datasets.

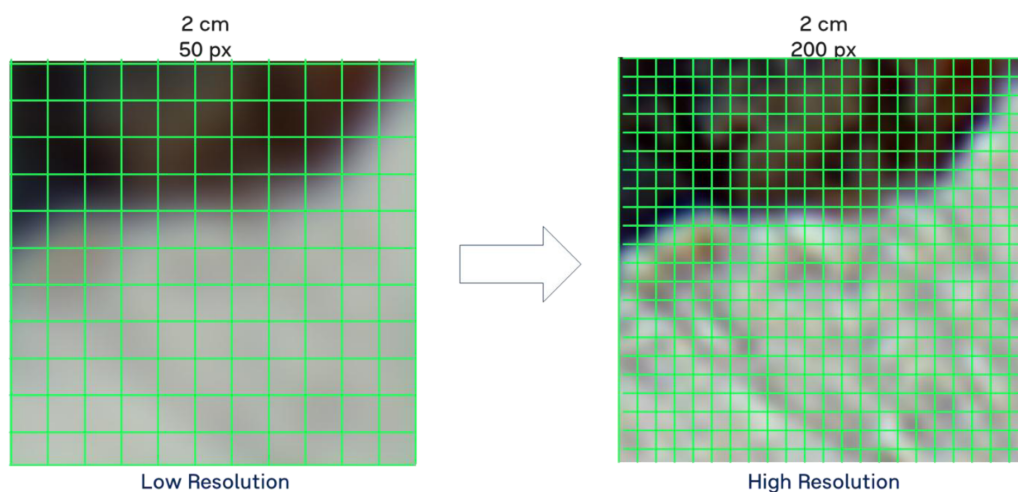


Figure 8—Resolution enhancement of a 2 cm wide core image from 50 pixels (low resolution) to 200 pixels (high resolution) using ESRGAN, showcasing improved detail and clarity.

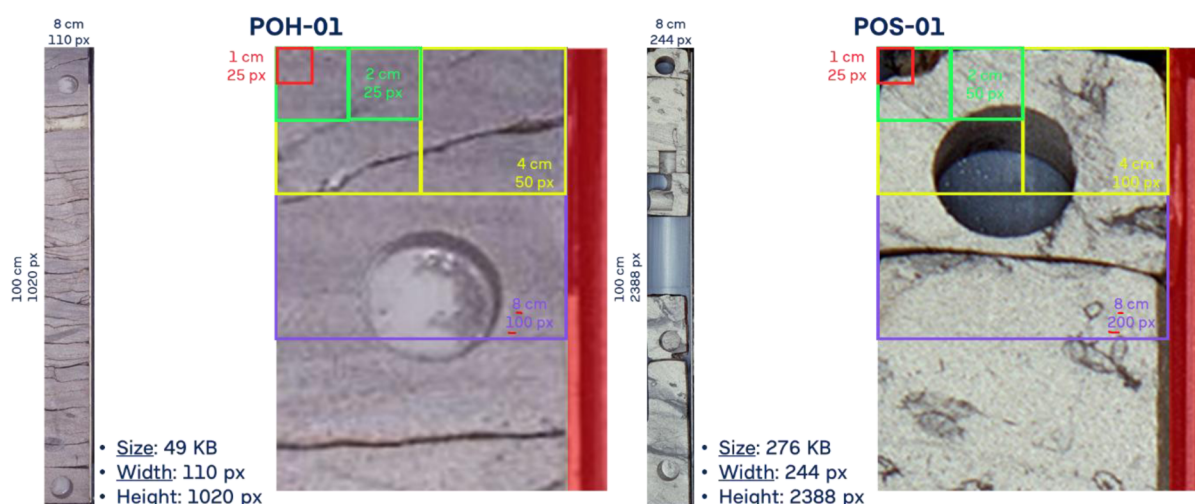


Figure 9—Core images from wells POH-01 and POS-01 before resolution enhancement. The POH-01 image measures 8 cm wide with 110 pixels and 1 m tall with 1020 pixels, while the POS-01 image has a higher resolution, measuring 8 cm wide with 244 pixels and 1 m tall with 2388 pixels.

Figure 10 illustrates the resolution enhancement of a core image from well POS-01 before and after applying the ESRGAN algorithm. The original low-resolution image on the left has a width of 8 cm with 200 pixels and a height of 1 m with 2300 pixels. After using ESRGAN, the image gains four times the pixel density in both dimensions, resulting in a high-resolution output with a width of 8 cm containing 800 pixels and a height of 1 m represented by 9200 pixels. This dramatic increase in resolution improves the image's detail and clarity, allowing for a more precise analysis of geological features. The transformation exemplifies the ability of ESRGAN to upscale images while preserving critical textures and patterns.

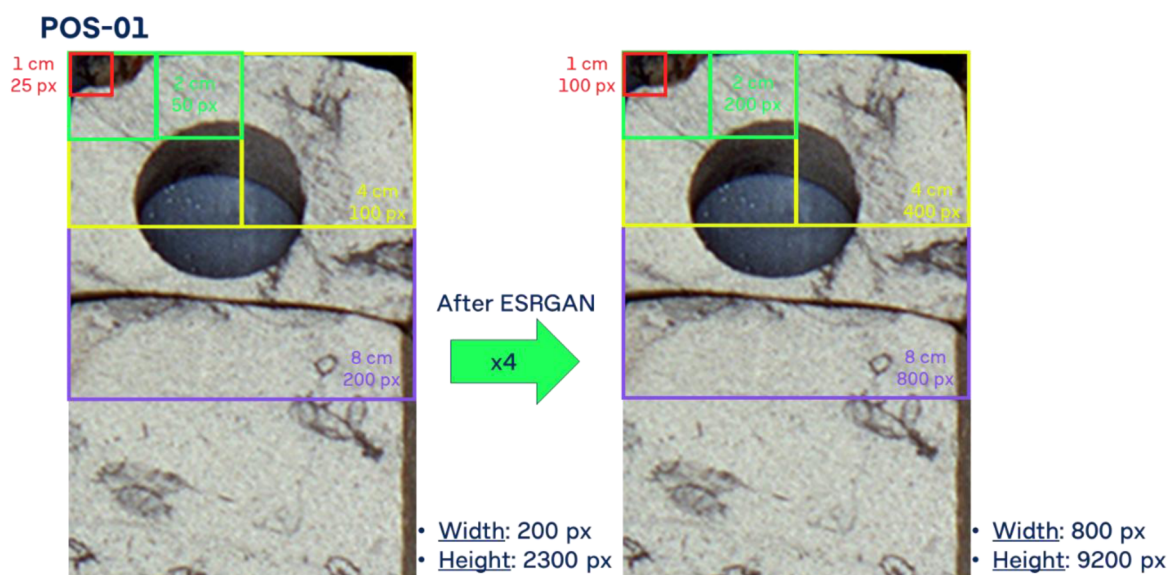


Figure 10—Resolution enhancement of a core image from well POS-01 using ESRGAN, increasing pixel density by 4x from 200 × 2300 pixels to 800 × 9200 pixels, significantly improving detail and clarity.

U-Net Image Segmentation

The enhanced white and UV images were segmented using a U-Net architecture. U-Net combines a contracting path for feature extraction with a symmetric expanding path for precise localization (Ronneberger et al., 2015). We adopted a patch-based approach: overlapping patches of 50×50 pixels were extracted from the ESRGAN outputs. Each patch contained three colour channels (RGB) for a single light

source or six channels when combining white and UV images. The dataset was split 70/30 for training and testing. The network was trained using categorical cross-entropy and the Adam optimizer. To reduce overfitting, we applied data augmentation (random flips and rotations) and regularization. Experiments showed that including both white and UV channels improved segmentation by providing complementary spectral information.

Image Labelling. The labelling method used for U-Net image segmentation relied on an Excel file (.csv) containing two key columns: Depth (m) and Lithofacies. The lithofacies column provided the lithofacies code for each depth interval. Using this reference file, each of the enhanced resolution core images was manually segmented based on the provided depth and lithofacies information.

The segmentation process assigned specific colors to represent each lithofacies class according to the scheme shown in Figure 11. The full classification and description of the lithofacies used in this segmentation process were detailed earlier in Section 3.1 (Facies Description). In addition to the lithofacies categories, four extra classes were defined and segmented: circular plug, lateral plug, artificial fractures (undefined), and preserved core. These additional categories were included to ensure comprehensive representation of all features observed in the core samples.

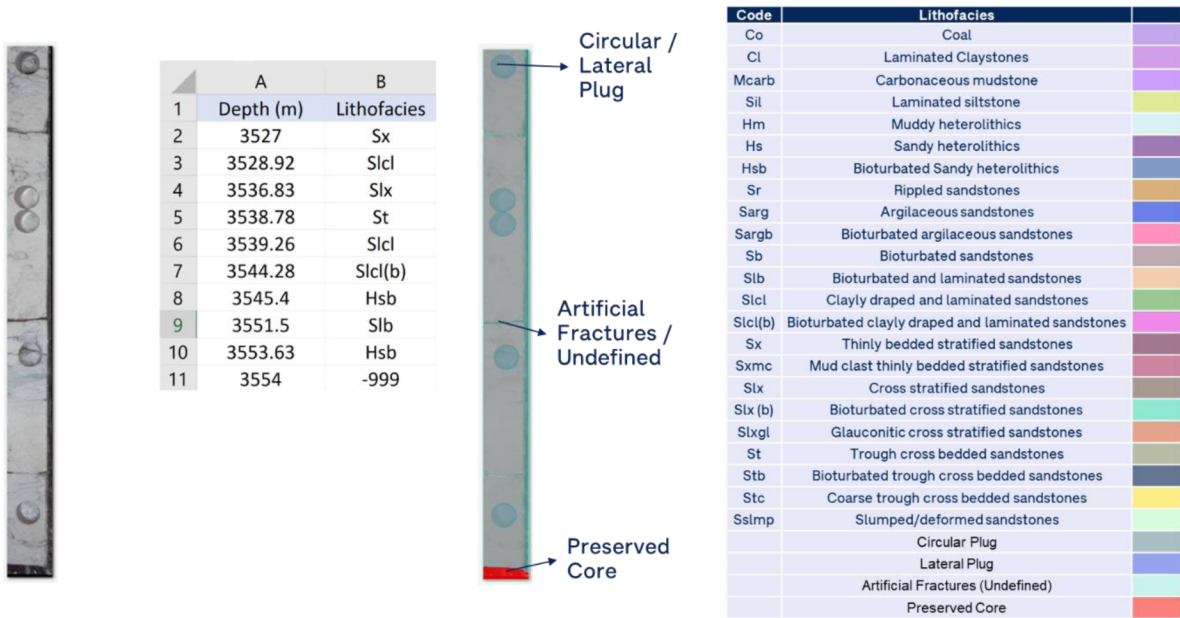


Figure 11—Manual segmentation of enhanced resolution core images using lithofacies codes from the Excel reference file and additional categories (circular plug, lateral plug, artificial fractures, preserved core), with colors assigned according to the lithofacies scheme described in Section 3.1.

By combining the lithofacies data with these supplementary categories, a complete pixel-wise annotation of the images was created, forming a robust dataset for training the U-Net model.

The U-Net architecture, widely used for image segmentation tasks, consists of a contracting path for feature extraction and an expansive path for upsampling and segmentation map generation. This architecture is designed to capture both spatial and contextual information by using convolutional layers, max-pooling operations, and skip connections that transfer fine details from the contracting to the expansive path.

In this case, a patch-based approach was used, where input patches of size 50x50x3 (RGB) were extracted from the enhanced resolution images to train the network. Figure 12 illustrates the process, showing how these patches were fed into the network for training, and the output segmentation map was evaluated using the previously described segmentation procedure. The U-Net was trained on these patches to distinguish features at the pixel level, achieving segmentation results with high accuracy.

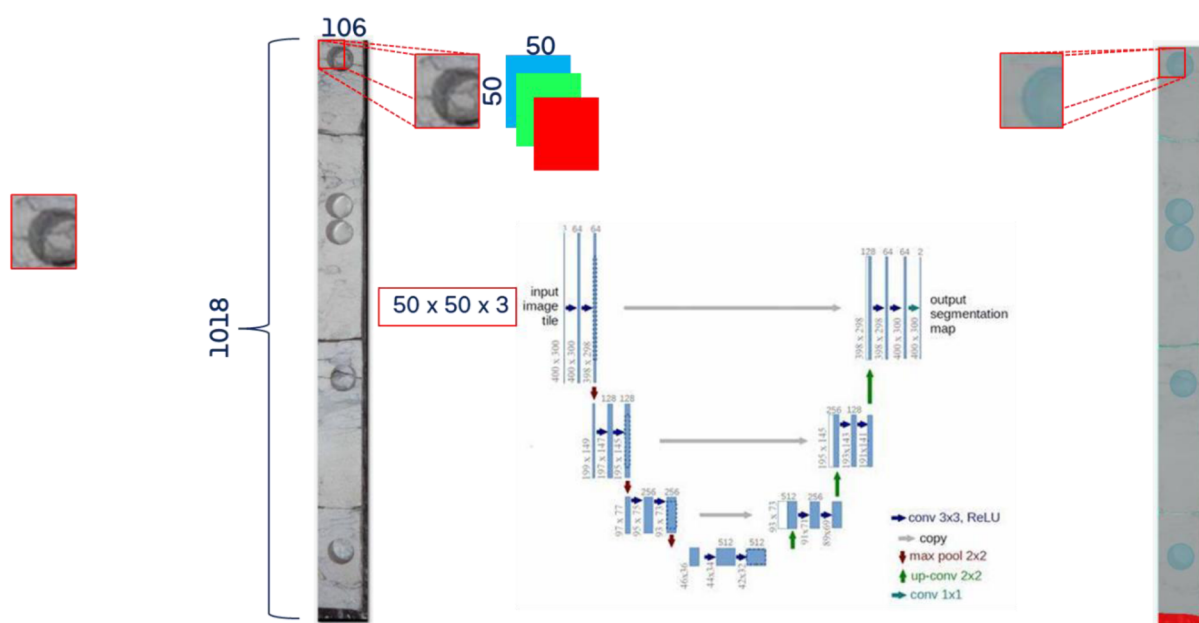


Figure 12—Training process for the U-Net architecture using 50x50x3 input patches, with segmentation map generation and evaluation for White and UV image datasets.

For training, 70% of the 1 m enhanced resolution White and UV images were used, while the remaining 30% was reserved for testing, ensuring a robust evaluation of the model. In another configuration, the White and UV images were combined as input channels, resulting in an input size of 50x50x6. This combined input configuration significantly improved the segmentation results, as it provided the network with additional information from both lighting conditions, enhancing its ability to differentiate features.

Figure 13 shows an example of patch selection. While the example highlights the patching strategy, the size and number of patches depicted are for reference purposes only. The actual segmentation process involved carefully selecting overlapping patches to maintain continuity and detail across the segmented images.



Figure 13—Example of patch selection from an image, with size and number of patches shown for reference purposes.

Figure 14 summarizes the number of patches generated from each well's enhanced resolution images for training and testing purposes in the ESRGAN workflow. On average, the images for POH-1 and POH-2

measured 400×4000 pixels, which, when divided into patches of 50×50 pixels, resulted in 640 patches per image. In the case of POS-1, each image measured 800×8200 pixels, yielding 2,640 patches per image. With 27 images from POH-1, 55 images from POH-2, and 94 images from POS-1, the total number of 50×50 patches across all wells amounted to 282,720 patches. These patches formed the dataset used for training and testing the network.

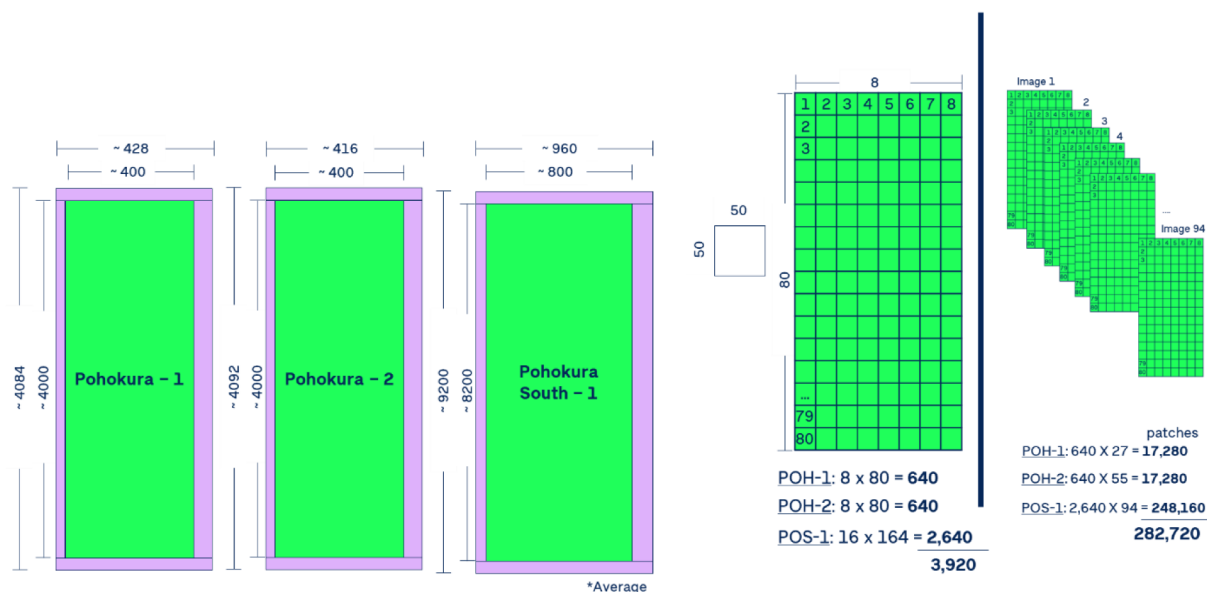


Figure 14—Total number of 50×50 patches generated from enhanced resolution images across the three wells, resulting in 282,720 patches for training and testing purposes.

Horizontal Voting for Facies

After segmentation, patches were reassembled to reconstruct each core segment. For each horizontal depth interval, we counted the number of pixels belonging to each facies and assigned the facies with the highest count to that interval. Majority voting is a simple yet effective ensemble technique that combines predictions from multiple classifiers to balance out individual weaknesses. In our case, the U-Net predictions across patches act as base classifiers whose outputs are aggregated by majority voting. This ensemble step converts noisy pixel-wise predictions into consistent depth-wise facies assignments.

Figure 15 demonstrates the voting process, where segmented patches from each depth are grouped and analyzed. The facies category with the most occurrences is then assigned to the corresponding depth interval, as illustrated by the majority column on the right.

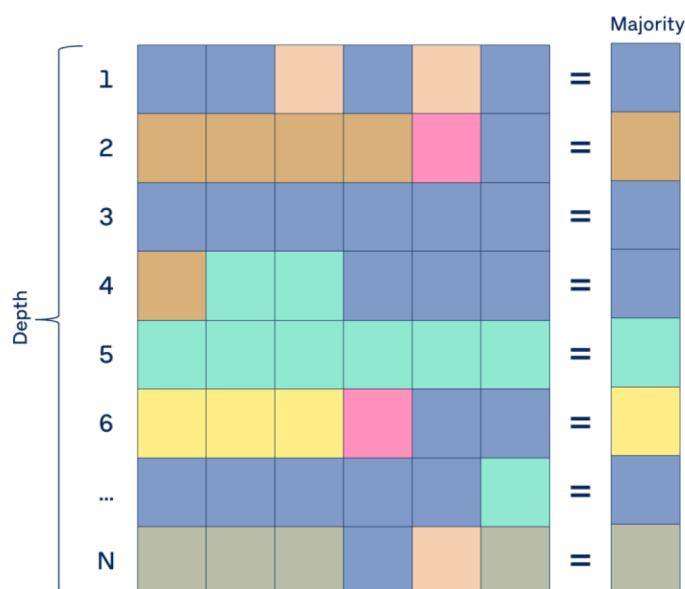


Figure 15—Voting process for assigning facies at each depth interval, based on the majority of segmented facies patches within the corresponding horizontal section.

Core Image Segmentation Results

ESRGAN training on 200 high-resolution Wisting images produced high-quality models. Applying the model to Pohokura images quadrupled the pixel density. Visual inspection showed that bedding planes, cross-laminations and bioturbated textures became clearer; figure captions should explicitly state the dimensions (e.g., the x-axis representing horizontal distance in centimetres and the y-axis representing depth in centimetres). The upscaling improved segmentation accuracy because the U-Net could identify features that were previously blurred. Comparisons between low-resolution and enhanced images demonstrated the benefits of super-resolution (Figure 10).

Figure 16 illustrates the segmentation results for the well POH-1, highlighting the combination of advanced methods used in this study for lithofacies classification. The first track displays the Gamma Ray (GR) log, which serves as a reference for identifying lithological changes across the depth intervals. The subsequent tracks provide additional context, beginning with the original lithofacies labels, which are based on manual annotations derived from the core descriptions. This is followed by the white light core image and the UV light core image, which capture different visual characteristics of the core, aiding in feature differentiation. The final track corresponds to the predicted lithofacies classifications, which are the output of the machine learning workflow.

The U-Net achieved a mean IoU of 0.68 on the test set for pixel-wise segmentation. When patches from both white and UV light were combined, the mean IoU increased to 0.72. After majority voting, the overall balanced accuracy for lithofacies classification was 0.77 with a macro-F1 score of 0.74. In comparison, Gjerazi (2023) study using Dataiku and limited training epochs achieved a balanced accuracy of approximately 0.60. The improvement is attributed not only to greater computational resources but mainly to the ESRGAN-based resolution enhancement, multi-channel inputs and deeper segmentation architecture. Confusion matrices revealed that previously under-represented facies (e.g., coal and carbonaceous mudstone) had higher recall, demonstrating better handling of class imbalance.

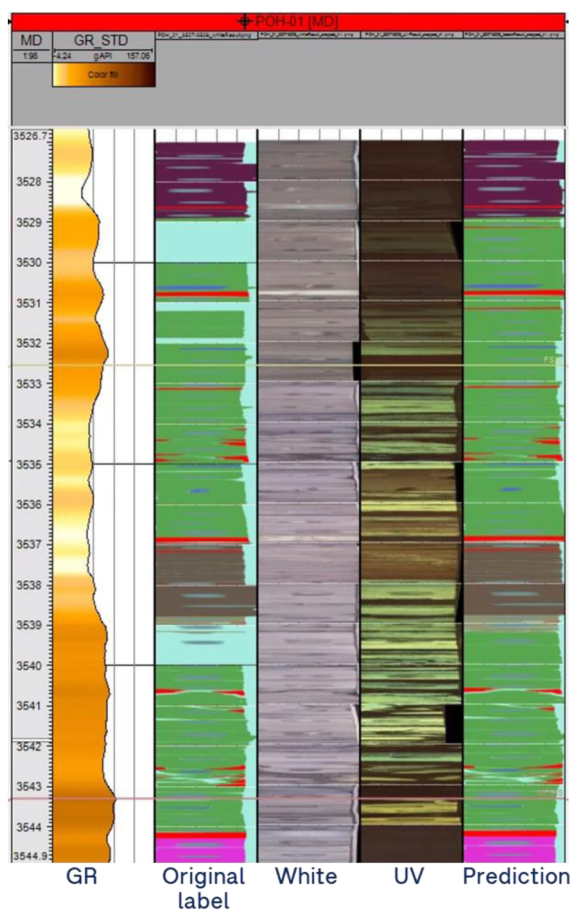


Figure 16—Segmentation results for POH-1 well showing GR log, original lithofacies label, white light core image, UV light core image, and predicted facies classification. Enhanced resolution, U-Net segmentation, and majority voting achieved a 77% facies prediction accuracy.

The dataset included images with different native resolutions: the POS-01 images had a resolution of $2,034 \times 2,660$ pixels whereas POH-01 and POH-02 images were roughly $884 \times 1,156$ pixels. Studies on medical imaging have shown that training convolutional neural networks with high-resolution images can improve classification for certain tasks, but multi-scale features may be more important than simply increasing resolution. In our case, upscaling the lower-resolution images enabled the model to learn consistent multi-scale features across wells. However, the POS-01 images still provided richer detail, contributing to slightly higher accuracy in those segments. Future work could explore multi-resolution inputs to further leverage this variability.

Histogram-based segmentation and K-Nearest Neighbours (K-NN) clustering were explored as baseline methods. Histogram segmentation partitioned images into colour-intensity ranges (red, green, blue and yellow) but produced coarse segments that did not correspond to geological features. K-NN clustering treated each pixel as a data point in RGB space and grouped pixels into k clusters using the elbow method to select k . Although K-NN produced more refined partitions, the clusters often corresponded to colour differences rather than facies, and classification accuracy remained low. Because these methods failed to capture context and texture, they were presented briefly for completeness and omitted from further analysis.

The combination of ESRGAN, U-Net segmentation and majority voting constitutes a reproducible and scalable workflow for lithofacies classification. ESRGAN eliminated the resolution disparity between wells, allowing the segmentation network to focus on geological features rather than artefacts. The U-Net's encoder-decoder architecture efficiently captured both global context and local details, while majority

voting aggregated pixel-wise predictions into robust depth-wise classifications. Balanced accuracy ensured fair evaluation across facies classes. Nevertheless, several challenges remain:

- Data imbalance – Despite improved performance, dominant facies still influenced the classifier. Techniques such as focal loss, oversampling and synthetic minority augmentation could be employed.
- Hyper-parameter tuning – ESRGAN hyper-parameters (e.g., patch size, number of residual blocks, training epochs) were adopted from the original paper; systematic tuning could yield further gains.
- Scalability – Applying this workflow to other geological settings will require training the ESRGAN on field-specific images to capture different textures. Wisting data provided adequate variety for the Pohokura field, but different lithologies may demand new training datasets.

Conclusions

This study demonstrates that super-resolution enhancement combined with deep learning and ensemble techniques markedly improves automated lithofacies classification. ESRGAN upscaling increased the spatial resolution of core photographs, revealing subtle sedimentary features. U-Net segmentation with six-channel inputs effectively delineated geological facies at the pixel level, and majority voting aggregated the results into consistent depth-wise facies assignments. The workflow achieved a balanced accuracy of 0.77 on the Pohokura dataset, outperforming previous approaches by integrating high-resolution imaging and multi-channel segmentation rather than simply increasing computational resources. Future work will address remaining class imbalance, tune hyper-parameters and evaluate transferability to other fields.

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